**1. Project Title: AI-Powered Food Waste Management Platform for Surplus Identification and Redistribution Optimization**

**2. Project Statement and Outcomes:**

Food service businesses, supermarkets, and households generate enormous quantities of avoidable food waste daily, driven by poor inventory management, inaccurate demand forecasting, and lack of systems to redirect surplus food to those who need it. This project builds a full-stack AI-powered food waste reduction web platform that enables food businesses to track inventory with real-time expiry monitoring, receive AI-driven waste risk predictions based on purchasing patterns and historical demand, and connect with NGOs and community kitchens to redistribute surplus food before it is wasted.

The deliverable is a dual-sided web platform serving food businesses and recipient organizations (NGOs, community kitchens, food banks). Business users gain an inventory management dashboard with waste risk scores, reorder recommendations, and surplus food listing tools. Recipient organizations access a live donation feed with pickup scheduling and logistics coordination. Analytics dashboards for both user types provide actionable insights on waste reduction trends, donation impact metrics, and food recovery volumes.

**3. Modules to be Implemented:**

1. Inventory Management & Expiry Tracking System
2. AI-Based Waste Prediction Engine
3. Redistribution Marketplace & Analytics Dashboard
4. System Integration, Testing & Project Finalization

**4. Week-wise Module Implementation and High-Level Requirements:**

**Weeks 1–2: Inventory Management & Expiry Tracking System**

- Build an inventory input system supporting manual entry, CSV bulk upload, and integration with POS systems via REST APIs.

Implement an expiry date tracking engine with automated alerts when items approach defined days-to-expiry thresholds.

- Develop a product categorization module using a food taxonomy to classify inventory items by perishability risk and storage requirements.
- Build barcode/QR scanning support for rapid inventory updates using a mobile-friendly web interface.
- Set up a multi-tenant database for storing business inventory records, transaction history, and surplus donation logs.

**Weeks 3–4: AI-Based Waste Prediction Engine**

- Collect and preprocess historical sales, purchase, and waste data to build time-series training datasets per product category.

- Train demand forecasting models (Prophet, LSTM) to predict daily and weekly demand and flag items at high waste risk.
- Build a waste risk scoring algorithm combining days-to-expiry, current stock level, and demand forecast into an actionable risk score.
- Implement a smart reorder recommendation module that suggests optimal purchase quantities based on demand forecasts and storage capacity.
- Deploy the waste prediction service as a FastAPI microservice with scheduled batch prediction runs and real-time on-demand scoring.

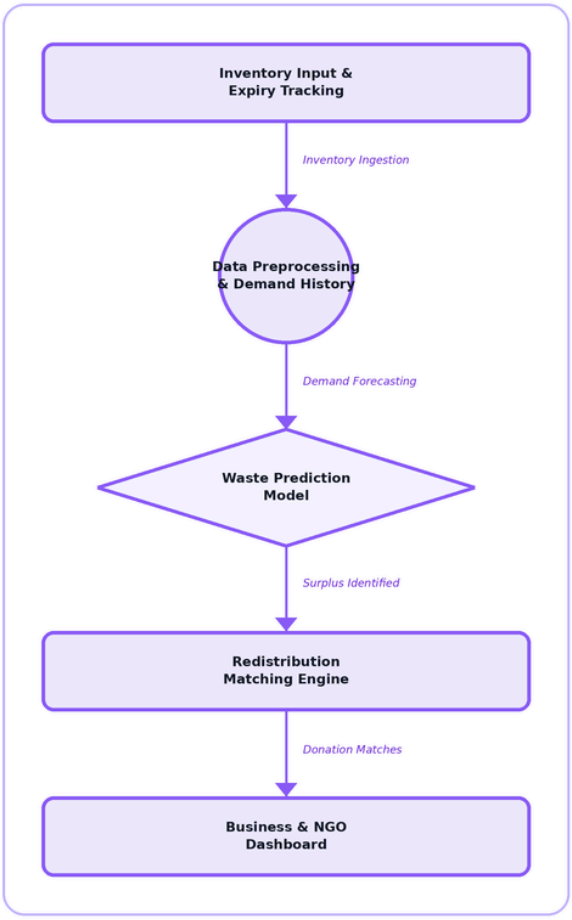
**Weeks 5–6: Redistribution Marketplace & Analytics Dashboard**

- Build a surplus food listing portal where businesses can post available items with quantity, expiry date, pickup location, and availability windows.
- Develop an AI-powered matching engine connecting surplus food listings with registered NGO requirements based on food type, quantity, and location proximity.
- Implement a pickup scheduling system with calendar-based coordination and confirmation notifications for both donors and recipients.
- Build React.js business and NGO dashboards: business view shows inventory health, waste risk charts, and donation tracker; NGO view shows incoming donations, pickup history, and beneficiary impact.
- Implement a sustainability impact dashboard showing aggregate food waste diverted, CO2 equivalents saved, and meals redistributed; deploy on cloud.

**Weeks 7–8: System Integration, Testing & Project Finalization**

- Conduct end-to-end integration testing across all modules — inventory input → waste predictor → redistribution matcher → business/NGO dashboard — and resolve any cross-module compatibility issues and edge-case failures.
- Perform load and stress testing on the backend APIs and cross-browser UI validation; address performance bottlenecks and implement enhancements from peer and mentor review.
- Refine the user experience based on testing feedback: improve interface clarity, add contextual help tooltips, and ensure full responsiveness on mobile and tablet devices.
- Prepare comprehensive project documentation covering system architecture, data flow diagrams, API reference, model or algorithm evaluation report, deployment guide, and known limitations.
- Develop the final project presentation (PPT) covering problem statement, proposed solution, architecture, key features, technology stack, results, challenges, and future scope — and rehearse the full end-to-end demo for a smooth, confident delivery.

**5. Flow Chart:**




## 6. Sample Dashboard:

FoodBridge
AI Food Waste Reduction Platform

Dashboard Listings NGOs Pickups Impact Settings

AP
Admin

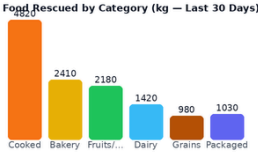
- Dashboard
- New Listing
- NGO Network
- Pickup Routes
- Impact Tracker
- Analytics
- Reports
- Settings

# Food Rescue Dashboard

Connecting surplus food to NGOs and communities in need

| Listings Active | Food Rescued | $CO_2$ Prevented | NGOs Active |
|---|---|---|---|
| 842 | 12,840 kg | 6,420 kg | 64 |
| 284 restaurants | this month | est. impact | serving 8,200 meals |

**Food Rescued by Category (kg — Last 30 Days)**




**Top Donor Partners** 30 days

| | |
|---|---|
| Hotel Sunshine | 1,840 kg |
| City Bakery | 980 kg |
| FreshMart Chain | 840 kg |
| Spice Garden Rest. | 720 kg |
| Campus Canteen | 640 kg |

**Recent Rescue Operations**

| Listing ID | Donor | Food Type | Qty (kg) | NGO Assigned | Status |
|---|---|---|---|---|---|
| LST-4821 | Hotel Sunshine | Cooked Meals | 48 | Hope Foundation | Picked Up |
| LST-4820 | City Bakery | Bread & Pastry | 32 | Annapoorna NGO | In Transit |
| LST-4819 | FreshMart | Fruits & Veg | 64 | Green Hands | Scheduled |
| LST-4818 | Campus Canteen | Rice & Curry | 28 | Seva Trust | Picked Up |
| LST-4817 | Spice Garden | Mixed Cooked | 42 | City Relief | Picked Up |